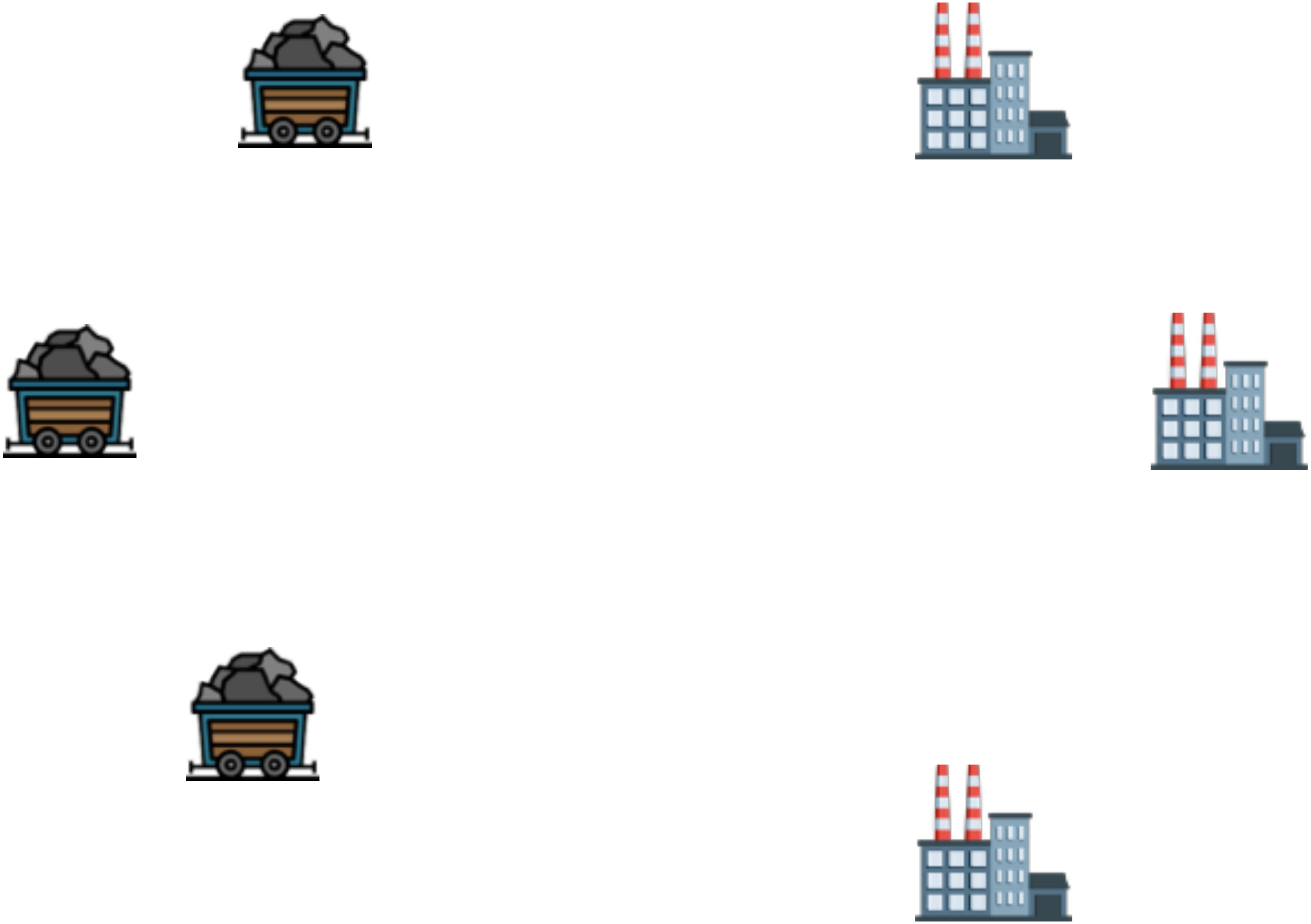
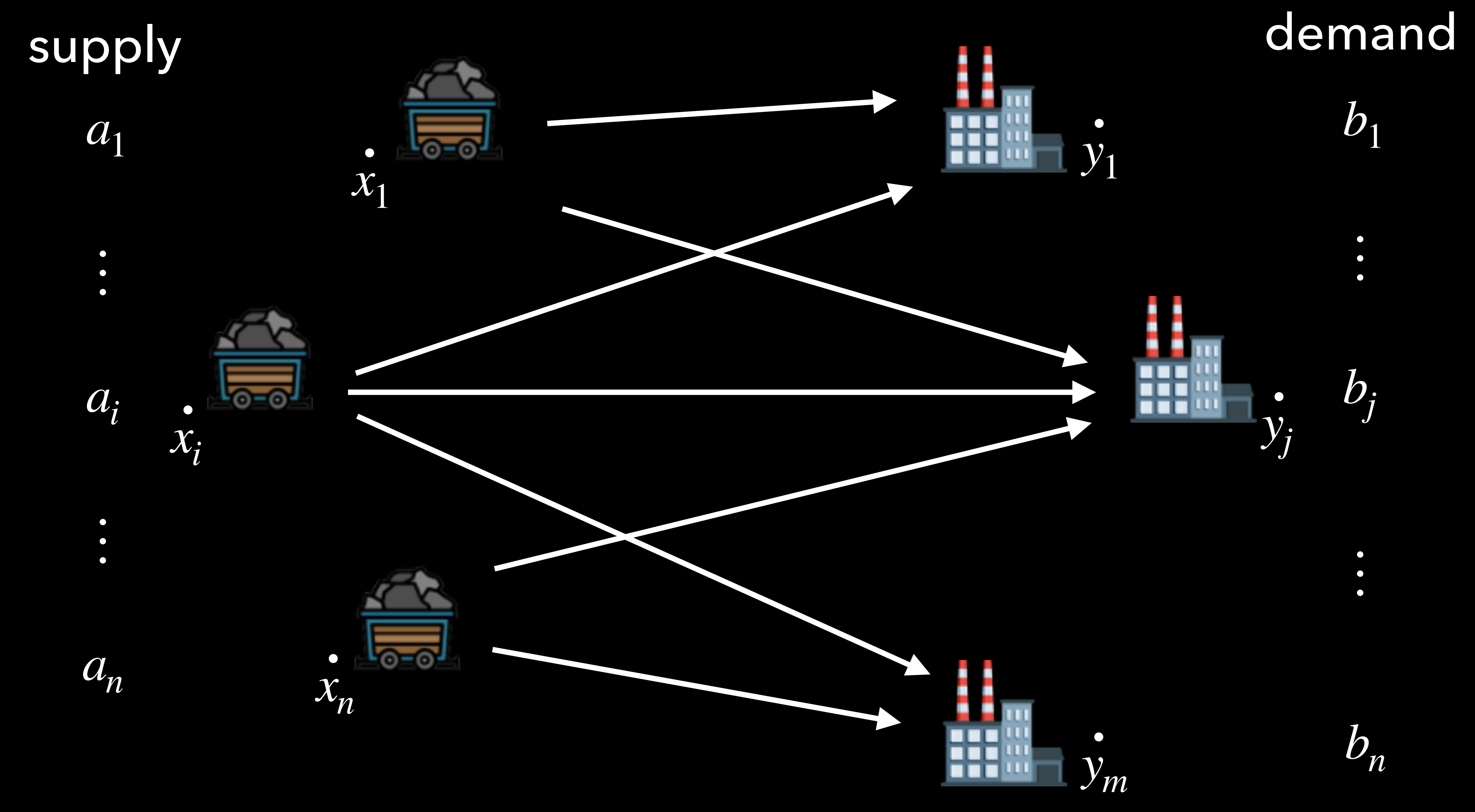


Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{C}_{i,j}\, \pi_{i,j}$$

Kantorovich Formulation

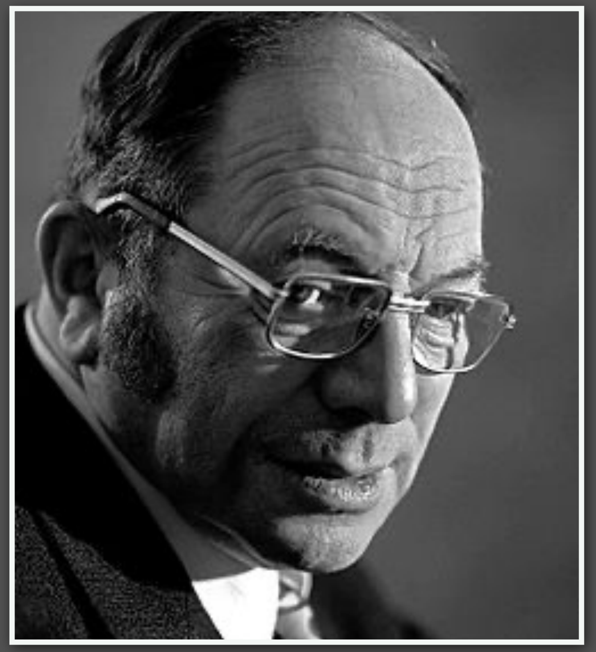


Cost matrix: $(C_{i,j})_{i,j=1}^{n,m} \in \mathbb{R}^{n \times m}$

Search over coupling matrices

$\Pi(\mathbf{a}, \mathbf{b}) = \{\pi \in \mathbb{R}^{n \times m} : \pi \geq 0, \pi \mathbf{1} = \mathbf{a}, \pi^\top \mathbf{1} = \mathbf{b}\}$

We want the min-cost one



Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{C}_{i,j} \, \pi_{i,j}$$

Kantorvich is an LP